FACEO-R: Functionally Anchored Overlapping Community Detection for Robust Pathway Module Discovery

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Motivation. Weighted gene co-expression network analysis (WGCNA) and its variants remain the dominant paradigm for partitioning genes into functional modules. We treat the validity of this paradigm as a statistical question rather than a modelling choice, and test it on a fixed benchmark with strictly paired comparisons and null models that separate genuine biological organisation from artefacts of network construction. Four assumptions are examined: that a monotone co-expression metric captures functional relatedness; that a global cut yields reproducible modules; that a single partition resolves nested sub-structure; and that reported modularity reflects biological organisation. All four fail.

Results. We assembled a benchmark of 16 publicly available expression cohorts spanning neoplastic and non-neoplastic disease, nine GO biological-process pathways and 31 partitioning algorithms from eight methodological families, yielding 100 pathway–cohort pairs and 3100 evaluation records. All algorithms operated on an identical topological overlap matrix per pair, enforcing a strictly paired comparison. Evaluation followed a six-layer protocol covering structural compactness, functional coherence (GO/KEGG/Reactome-referenced, size-unbiased z-scores), bootstrap stability, multi-perturbation robustness, biological validity (curated antagonistic gene pairs) and downstream clinical prediction. To eliminate circularity, graph representation learning was reformulated as self-supervised edge prediction, removing all community labels from training. Synthetic benchmarks with known ground truth supplied absolute accuracy references. Motivated by the finding that expression graphs degrade under realistic constraints, we further introduce **FACEO-R**, a functionally anchored overlapping community detection method in which the weight of GO-derived functional similarity adapts to the estimated reliability of the expression graph. Under controlled degradation (sample subsampling, expression noise and feature dropout) and on clean data alike, FACEO-R outperformed Leiden at $\gamma = 1.0$ in 118/145 comparisons (81%; paired Wilcoxon $p = 1.9 \times 10^{-14}$) and WGCNA-style hierarchical cutting in 127/145 (88%; $p = 9.0 \times 10^{-20}$), with the largest gains under feature dropout ($\Delta = +0.070$, 27/28 wins). Multi-resolution community detection produced functionally more coherent modules than hierarchical clustering (median $z = 0.689$ versus 0.045; paired Wilcoxon $q = 4.6 \times 10^{-7}$) while simultaneously improving modularity (0.291 versus 0.000; $q = 2.0 \times 10^{-17}$), and recovered nested functional sub-structures with markedly higher accuracy in synthetic benchmarks (ARI 1.000 versus 0.908). Downstream clinical prediction also favoured multi-resolution partitions (AUC 0.989 versus 0.898).

Conclusions. We find that the three implicit assumptions of current practice — a single adequate algorithm, a single defensible resolution, and modularity as evidence — all fail, and that the failures are systematic. First, modularity — and its significance against degree-preserving null models — proved unstable with respect to graph construction (threshold sensitivity reverses the sign; weights contribute 29%), so it should not be used as primary evidence. Second, the functionally optimal resolution is *not predictable*: optimal γ was highly dispersed across pathways (entropy 1.74 of a maximum 1.79, with no value optimal in more than 26% of cases), yet four families of automatic selection strategies — structural-fingerprint regression, pathway-size grouping, multi-resolution consensus clustering and cross-library held-out annotation guidance — all failed (accuracy 0.16–0.46, none significantly better than chance, with consensus clustering actively harmful). Third, any fixed resolution therefore sacrifices a mean of 0.765 functional-coherence z-units relative to per-pathway optimal selection. We recommend reporting resolution-sensitivity curves validated against held-out functional evidence, rather than modularity, module counts, or a single arbitrary resolution.

Availability and implementation: The FACEO-R implementation, all analysis scripts, processed expression matrices and complete evaluation tables are freely available at https://github.com/sunmy1129/faceo-r under an MIT licence.
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Keywords: functional module partitioning; community detection; Leiden algorithm; WGCNA; graph neural networks; self-supervised learning; benchmarking

Appendix

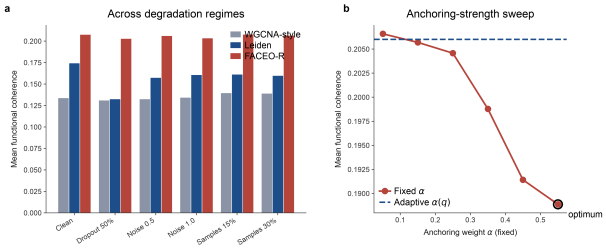


Fig. S1. The advantage is present with and without degradation, and the anchoring strength has a narrow optimum. (a) Mean functional coherence under clean data and five degradation regimes; FACEO-R exceeds both baselines in every condition. (b) Coherence as a function of a fixed anchoring weight α (solid); performance improves monotonically as anchoring strengthens, the originally proposed adaptive schedule under-anchored (dashed, below every fixed setting in $[0.05, 0.35]$), and the retuned schedule matches the best fixed weight.

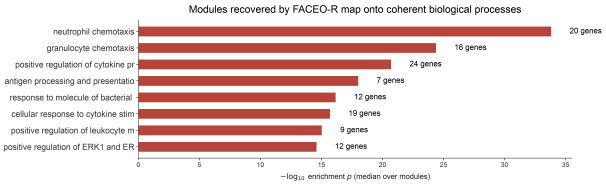


Fig. S2. Recovered modules map onto coherent, specific biological processes. Median enrichment significance of the top Gene Ontology term per module, with the input pathway excluded from consideration; annotations show the median number of module genes in each term. Twenty-two distinct processes were recovered across four cohorts.